Ch. 5.1-5.2 HW Solns

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QuickQuiz

2. (d) – An increase in price leads to a decrease in total revenue if demand is elastic (with respect to price).

3. (d) – The elasticity of a linear demand curve is non-constant.

4. (c) – If the income elasticity of demand is 0.5, the decrease in income between Gondor and Rohan leads to a proportionately smaller decrease in Q_D. Since Rohan's income declines more than its quantity demanded relative to Gondor, Rohan must spend a higher fraction of its income on food.

Problems and Applications

2. (a) i.

$$\%\Delta Q = \frac{1900 - 2000}{\frac{1900+2000}{2}} = \frac{-100}{1950} = -\frac{2}{39}$$

$$\%\Delta P = \frac{250 - 200}{\frac{250+200}{2}} = \frac{50}{225} = \frac{2}{9}$$

$$\eta = \left|\frac{\%\Delta Q}{\%\Delta P}\right| = \frac{\frac{2}{39}}{\frac{2}{9}} = \frac{9}{39} \approx 0.23$$

ii.

$$\%\Delta Q = \frac{600 - 800}{\frac{600+800}{2}} = \frac{-200}{700} = \frac{-2}{7}$$

$$\%\Delta P = \frac{2}{9} \text{ (See above)}$$

$$\eta = \left|\frac{\%\Delta Q}{\%\Delta P}\right| = \frac{\frac{2}{7}}{\frac{2}{9}} = \frac{9}{7} \approx 1.29$$

(b) Flights are more of a necessity for business travelers and more of a luxury for vacationers, so we would expect vacationers to have more elastic demand.

3. (a) Short run:

$$\%\Delta P = \frac{2.2 - 1.8}{\frac{2.2+1.8}{2}} = \frac{0.4}{2} = 0.2\%$$
$$\eta_{SR} = \left|\frac{\%\Delta Q_{SR}}{\%\Delta P}\right|$$
$$0.2 = \frac{|\%\Delta Q_{SR}|}{0.2}$$
$$0.04 = |\%\Delta Q_{SR}|$$
$$-0.04 = \%\Delta Q_{SR} \text{ (by the law of demand)}$$

In the short run, the quantity of heating oil demanded decreases by 0.04%.

Long run:

$$\eta_{LR} = \left|\frac{\%\Delta Q_{LR}}{\%\Delta P}\right|$$
$$0.7 = \frac{|\%\Delta Q_{LR}|}{0.2}$$
$$0.14 = |\%\Delta Q_{LR}|$$
$$-0.14 = \%\Delta Q_{LR} \text{ (by the law of demand)}$$

In the long run, the quantity of heating oil demanded decreases by 0.14%.

(b) A longer time horizon gives people more time to find substitutes for heating oil and change their behavior – for example switching their home to natural gas – so we would expect demand to be more elastic in the long run.

4. If total revenue increases, demand is inelastic because Q_D must decrease proportionately less than P increases.

5. (a) i.

$$\%\Delta Q = \frac{32 - 40}{\frac{32+40}{2}} = -\frac{8}{36} = -\frac{2}{9}$$
$$\%\Delta P = \frac{10 - 8}{\frac{10+8}{2}} = \frac{2}{9}$$
$$\eta = \left|\frac{\%\Delta Q}{\%\Delta P}\right| = 1$$

ii.

$$\%\Delta Q = \frac{45-50}{\frac{45+50}{2}} = \frac{-5}{47.5} = \frac{-10}{95}$$

$$\%\Delta P = \frac{2}{9}$$

$$\eta = \left|\frac{\%\Delta Q}{\%\Delta P}\right| = \frac{90}{190} = \frac{9}{19} \approx 0.47$$

(b) i.

$$\%\Delta Q = \frac{30-24}{\frac{30+24}{2}} = \frac{6}{27} = \frac{2}{9}$$

$$\%\Delta I = \frac{24,000-20,000}{\frac{24,000+20,000}{2}} = \frac{4,000}{22,000} = \frac{2}{11}$$

$$IED = \frac{\%\Delta Q}{\%\Delta I} = \frac{\frac{2}{9}}{\frac{2}{11}} = \frac{11}{9} \approx= 1.22$$

ii.

$$\%\Delta Q = \frac{12-8}{\frac{12+8}{2}} = \frac{4}{10} = \frac{2}{5}$$

$$\%\Delta I = \frac{2}{11}$$

$$IED = \frac{\%\Delta Q}{\%\Delta I} = \frac{\frac{2}{5}}{\frac{2}{11}} = \frac{11}{5} = 2.2$$